

Determining Voxel ID of Detector A from McStas Simulation Event Positions

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Parameters

x_e, y_e, z_e is the position of events in the right-handed Cartesian coordinate system in which axis Z is along the incident beam and the axis Y is the vertical direction opposite to gravity (it should be perpendicular to Z). The coordinate system is centred on the sample.

γ_d is the gamma of the detector A.

r_d is the distance from the sample to the active area of the detector A that is equal to 1 m.

$\Delta\gamma_{v.s.}$ is the angle between two vertical segments that is equal to 1.05° .

$\Omega_{v.s.}$ is the angle of inclination of the vertical segment that is equal to -6.065° .

$r_{v.s.}$ is the depth of the vertical segment that is equal to 0.530 m.

a_t, b_t, a_b, b_b are parameters describing the height of vertical segments. They are equal to 252.863 mm., 416.545 mm., -722.903 mm., -1077.999 mm., respectively.

$N_c, N_a, N_{v.s.}$ is the total number of voxels along cathode (32 - depth), anode (128 - vertical) and vertical segments (60 - horizontal).

$n_c, n_a, n_{v.s.}$ is the number of cathode/anode/vertical segment for the current event.

ID is the identification number of the detector.

Expressions

Cathode position:

$$x_c = \frac{\sqrt{x_e^2 + z_e^2 - r_d^2 \cos^2 \Omega_{v.s.}} - r_d \cos \Omega_{v.s.}}{r_{v.s.}}$$

$$n_c = \text{floor}(N_c \cdot x_c)$$

Anode position:

$$a'_t = a_t + (b_t - a_t) \cdot x_c$$

$$a'_b = a_b + (b_b - a_b) \cdot x_c$$

$$y_a = \frac{y_e - a'_t}{a'_b - a'_t}$$

$$n_a = \text{floor} (N_a \cdot y_a)$$

Vertical segment position:

$$\tan \gamma = \frac{x_e}{z_e}$$

$$\sin \Delta\gamma' = \sin (\Omega_{\text{v.s.}}) \cdot x_c \frac{r_{\text{vs}}}{\sqrt{x_e^2 + z_e^2}}$$

$$z_{\text{v.s.}} = \frac{\gamma - \gamma_d - \Delta\gamma' + \Delta\gamma_{\text{v.s.}}/2}{N_{\text{v.s.}} \Delta\gamma_{\text{v.s.}}}$$

$$n_{\text{v.s.}} = \text{floor} (2N_{\text{v.s.}} \cdot z_{\text{v.s.}})$$

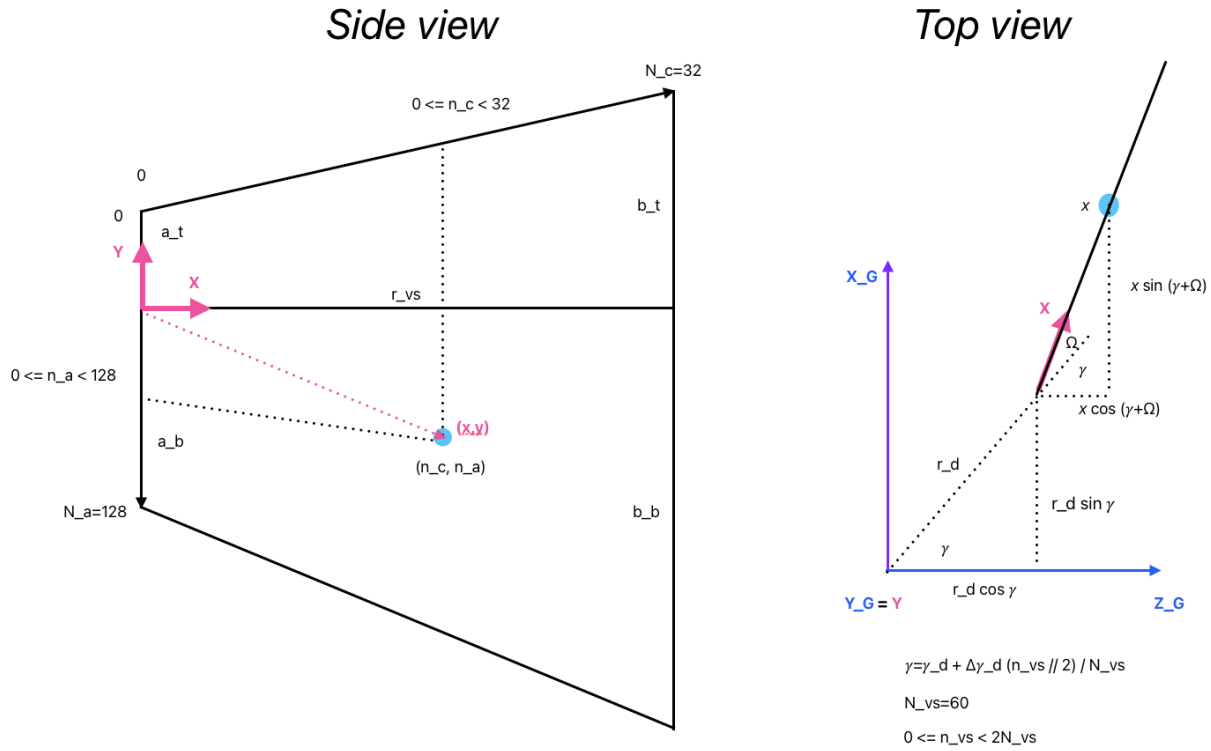
In the expression above, N_{vs} is doubled as every vertical segment has two voxels (from each side of the vertical segment).

The identification number of a voxel for detector A is therefore calculated as

$$\text{ID} = n_{\text{v.s.}} + (n_a + n_c \cdot N_a) \cdot 2N_{\text{v.s.}} + 1$$

Estimation of position of event based on ID of voxel

Vertical segment



Relation between n-a, n_c, n_vs and x_G, y_G, z_G

	Px	Pz (Py for CDT)
v_1	59,448	1526,823
v_2	32,937	1529,143
v_3	8,418	999,264
v_4	-8,418	1000,736
c_end	46,193	1527,983
c_origin	0,000	1000,000
Length	530,000	mm.
Ω_{vs}	5,000	deg.

Estimation of position and length of active area of Detector A from voxel map provided by CDT.

	X	Y	Z
Front	0,00000	252,86300	1000,00000
End	46,19250	416,54500	1527,98300
a_t:	252,86300		
b_t	416,54500		
Front	0,00000	-722,90300	1000,00000
End	46,19250	-1077,99900	1527,98300
a_b:	-722,90300		
b_b:	-1077,99900		

Values a_t, b_t, a_b, b_b based on the voxel map sent by CDT

$$x_c = \frac{n_c + 1/2}{N_c}$$

$$y_a = \frac{n_a + 1/2}{N_a}$$

$$x = r_{v.s.} \cdot x_c$$

$$a'_t = a_t + (b_t - a_t) \cdot x_c$$

$$a'_b = a_b + (b_b - a_b) \cdot x_c$$

$$y = a'_t + (a'_b - a'_t) \cdot y_a$$

$$z_{v.s.} = \frac{n_{v.s.} // 2}{N_{v.s.}}$$

$$\gamma = \gamma_d + z_{v.s.} \cdot \Delta\gamma_{v.s.} \cdot N_{v.s.}$$

$$x_e = r_d \cdot \sin \gamma + x \cdot \sin (\gamma + \Omega_{v.s.})$$

$$z_e = r_d \cdot \cos \gamma + x \cdot \cos (\gamma + \Omega_{v.s.})$$

$$y_e = y$$